

Architecture des ordinateurs

TP 1

Circuits logiques.

→ Exercice 1 :

b) Non $\rightarrow \bar{X} = \overline{X \cdot X} = X \uparrow X$

Oui $\rightarrow X = \overline{\bar{X}} = \overline{\overline{X \cdot X}} = \overline{X \uparrow X} = (X \uparrow X) \uparrow (X \uparrow X)$

Et $\rightarrow X \cdot Y = \overline{\overline{X \cdot Y}} = \overline{X \uparrow Y} = (X \uparrow Y) \uparrow (X \uparrow Y)$

Ou $\rightarrow X + Y = \overline{\overline{X + Y}} = \overline{\bar{X} \cdot \bar{Y}} = \overline{X \uparrow X} \cdot \overline{Y \uparrow Y} = (X \uparrow X) \uparrow (Y \uparrow Y)$

c) Non $\rightarrow \bar{X} = \overline{X + X} = X \downarrow X$

Oui $\rightarrow X =$

Et $\rightarrow X \cdot Y = \overline{\overline{X \cdot Y}} = \overline{\overline{X + Y}} = \overline{X \downarrow Y} = (X \downarrow Y) \downarrow (X \downarrow Y)$

Ou $\rightarrow X + Y = \overline{\overline{X + Y}} = \overline{X \downarrow Y} = (X \downarrow Y) \downarrow (X \downarrow Y)$

e.d) a b x $\Rightarrow x = \bar{a}b + a\bar{b} = a \oplus b$

a	b	x
0	0	0
0	1	1
1	0	1
1	1	0

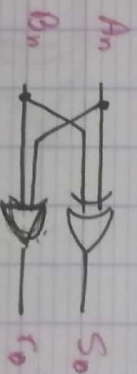


→ Exercice 2 :

1) Demi-additionneur :

A_0	B_0	S_0	r_0
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$\Rightarrow \begin{cases} S_0 = \bar{A}_0 \cdot B_0 + A_0 \cdot \bar{B}_0 = A_0 \oplus B_0 \\ r_0 = A_0 \cdot B_0 \end{cases}$



1) $AD = 100$

12

2) Additionner Complet :

A_n	B_n	r_{n-1}	S_n	r_n
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

→ Methode de Karnaugh :

$r_{n-1} / A_n B_n$	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$\Rightarrow S_n = \bar{A}_n \cdot \bar{B}_n \cdot \bar{r}_{n-1} + \bar{A}_n \cdot B_n \cdot \bar{r}_{n-1} + A_n \cdot \bar{B}_n \cdot r_{n-1} + A_n \cdot B_n \cdot r_{n-1}$$

$$= A_n \oplus B_n \oplus r_{n-1}$$

$$\Rightarrow S_n = \bar{A}_n \cdot \bar{B}_n \cdot \bar{r}_{n-1} + \bar{A}_n \cdot B_n \cdot \bar{r}_{n-1} + A_n \cdot \bar{B}_n \cdot r_{n-1} + A_n \cdot B_n \cdot r_{n-1}$$

$$= \bar{A}_n \cdot (\bar{B}_n \oplus r_{n-1}) + A_n \cdot (B_n \oplus r_{n-1})$$

$$= A_n \oplus B_n \oplus r_{n-1}$$

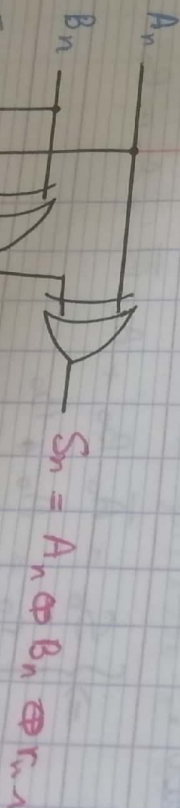
$$\Rightarrow r_n = \bar{A}_n \cdot B_n \cdot r_{n-1} + A_n \cdot \bar{B}_n \cdot r_{n-1} + A_n \cdot B_n \cdot \bar{r}_{n-1} + A_n \cdot B_n \cdot r_{n-1}$$

$$= r_{n-1} \cdot (A_n \oplus B_n) + A_n \cdot B_n$$

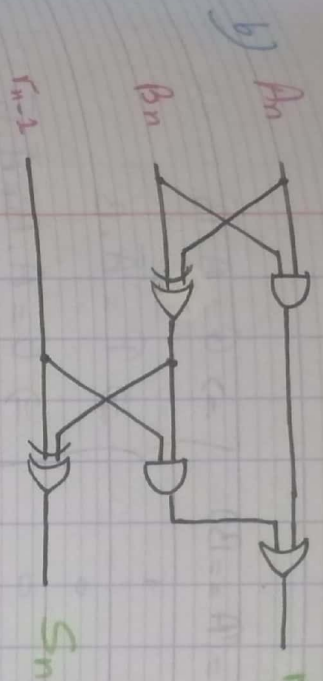
$r_{n-1} / A_n B_n$	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$\Rightarrow r_n = \bar{A}_n \cdot B_n \cdot r_{n-1} + A_n \cdot \bar{B}_n \cdot r_{n-1} + A_n \cdot B_n \cdot \bar{r}_{n-1} + A_n \cdot B_n \cdot r_{n-1}$$

$$= r_{n-1} \cdot (A_n \oplus B_n) + A_n \cdot B_n$$

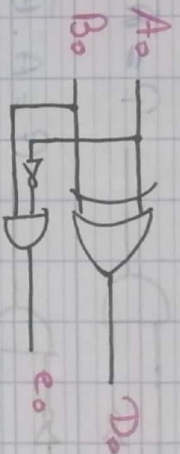


$$r_n = r_{n-1} \cdot (A_n \oplus B_n) + A_n \cdot B_n$$



3) Soustracteur - demi :

$$A_0 \quad B_0 \quad D_0 \quad e_0 \Rightarrow \begin{cases} D_0 = \overline{A_0} \cdot B_0 + A_0 \cdot \overline{B_0} = A_0 \oplus B_0 \\ e_0 = \overline{A_0} \cdot B_0 \end{cases}$$



4) Soustracteur Complet :

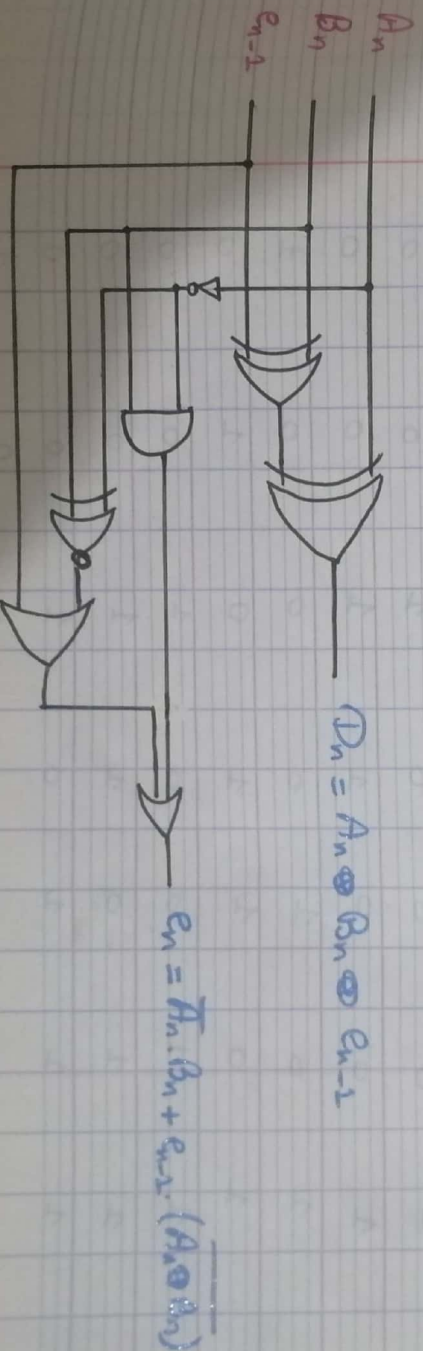
A_n	B_n	e_{n-1}	D_n	e_n
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	0
1	1	1	0	0

$$\Rightarrow D_n = \overline{A_n} \cdot \overline{B_n} \cdot \overline{e_{n-1}} + \overline{A_n} \cdot B_n \cdot \overline{e_{n-1}} + A_n \cdot \overline{B_n} \cdot \overline{e_{n-1}} + A_n \cdot B_n \cdot e_{n-1}$$

$$= A_n \oplus B_n \oplus e_{n-1}$$

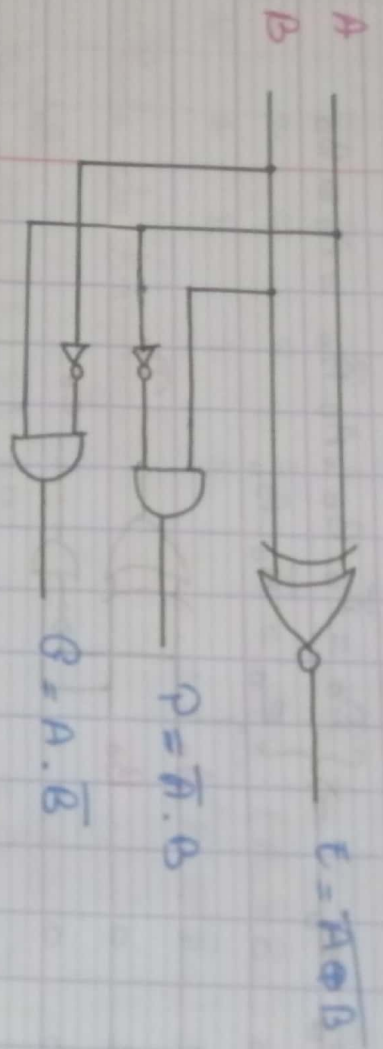
$$\Rightarrow e_n = \overline{A_n} \cdot \overline{B_n} \cdot e_{n-1} + \overline{A_n} \cdot B_n \cdot \overline{e_{n-1}} + A_n \cdot \overline{B_n} \cdot e_{n-1} + A_n \cdot B_n \cdot \overline{e_{n-1}}$$

$$= \overline{A_n} \cdot B_n + e_{n-1} \cdot (\overline{A_n} \oplus B_n)$$



5) Comparateur élémentaire :

A	B	$G = A > B$	$P = A < B$	$E = (A = B)$	
0	0	0	0	1	$\Rightarrow G = A \cdot \bar{B}$
0	1	0	1	0	$\Rightarrow P = \bar{A} \cdot B$
1	0	1	0	0	$\Rightarrow E = A \cdot B + \bar{A} \cdot \bar{B}$
1	1	0	0	1	$= A \oplus B$



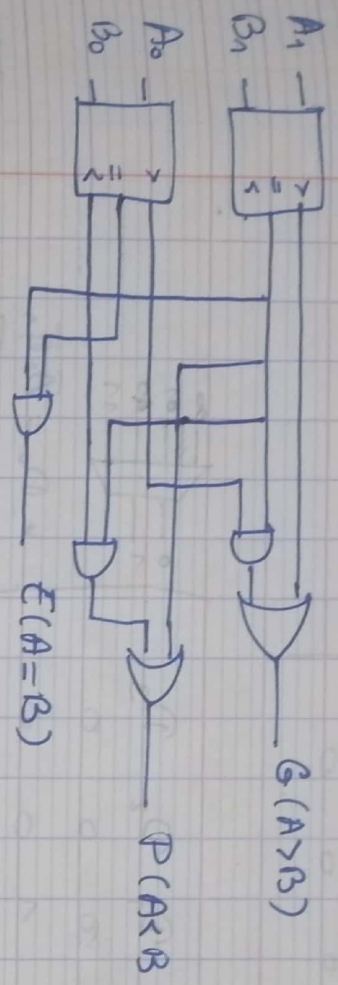
6) Comparateur Complet :

A_1	A_0	B_1	B_0	G	P	E
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	1	0	0
0	0	1	1	0	0	1
0	1	0	0	0	1	0
0	1	0	1	0	0	0
0	1	1	0	1	0	0
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	0	1	0
1	0	1	0	0	0	1
1	0	1	1	1	0	0
1	1	0	0	0	1	0
1	1	0	1	0	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

$G = (A_1 > B_1) \text{ OR } (A_1 = B_1 \text{ AND } A_0 > B_0)$
 $= A_1 \overline{B_1} + (\overline{A_1 \oplus B_1}) \cdot A_0 \overline{B_0}$

$P = (\overline{A_1} B_1 + (\overline{A_1 \oplus B_1}) \cdot \overline{A_0} B_0$

$E = (A_1 = B_1) \text{ AND } (A_0 = B_0) = \overline{A_1 \oplus B_1} \cdot \overline{A_0 \oplus B_0}$

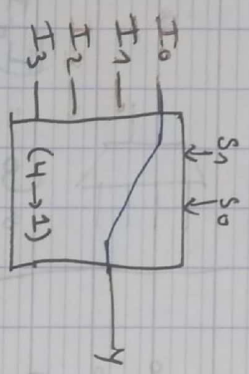


4/ Mux:

a) $S_1 \quad S_0 \quad Y$

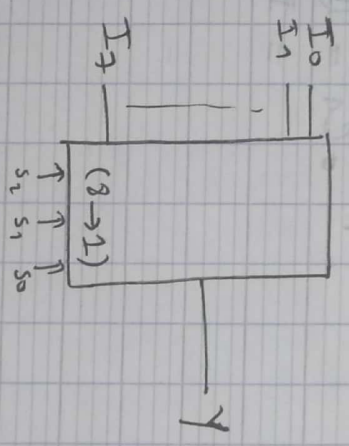
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

$Y = I_0 \cdot (\overline{S_1} \overline{S_0}) + I_1 \cdot (\overline{S_1} S_0) + I_2 \cdot (S_1 \overline{S_0}) + I_3 \cdot (S_1 S_0)$



b) $S_2 \quad S_1 \quad S_0 \quad Y$

0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

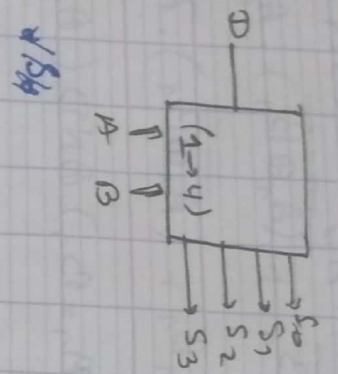


$Y = E \cdot \Sigma(m_i)$

8x1

2/ Demux :

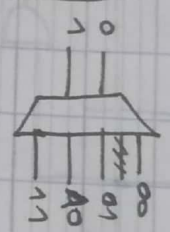
A_1	A_0	S_3	S_2	S_1	S_0
0	0	0	0	0	0
0	1	0	0	1	0
1	0	1	0	0	0
1	1	1	1	0	0



$$\begin{aligned} S_0 &= \overline{A_1} \overline{A_0} \\ S_1 &= \overline{A_1} A_0 \\ S_2 &= A_1 \overline{A_0} \\ S_3 &= A_1 A_0 \end{aligned}$$

3/ Multiplexer :

A_1	A_0	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

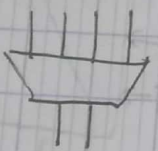


$$\begin{aligned} D_0 &= \overline{A_0} \overline{A_1} \\ D_1 &= \overline{A_0} A_1 \\ D_2 &= A_0 \overline{A_1} \\ D_3 &= A_0 A_1 \end{aligned}$$

$$D_3 = A_0 A_1$$

10/ Expression :

I_0	I_1	I_2	I_3	C_1	C_0
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1



$$\begin{aligned} C_0 &= \overline{I_0} \cdot \overline{I_1} \cdot \overline{I_2} \cdot \overline{I_3} + \overline{I_0} \cdot \overline{I_1} \cdot \overline{I_2} \cdot I_3 \\ C_1 &= \overline{I_0} \cdot \overline{I_1} \cdot I_2 \cdot \overline{I_3} + \overline{I_0} \cdot \overline{I_1} \cdot I_2 \cdot I_3 \\ &= \overline{I_0} \cdot \overline{I_1} \cdot (I_2 \oplus I_3) \end{aligned}$$